

*Ab Initio Whole Cell Kinetic Model of Chromobacterium violaceum* ATCC 12472 (cviFA26)

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Abstract

Chromobacterium violaceum ATCC 12472, a Gram-negative bacterium, is primarily found in soil and freshwater, mainly in tropical or subtropical regions, thriving in warm climates. Despite being an opportunistic pathogen, *C. violaceum* ATCC 12472 is known to have therapeutic potential as well other useful qualities. It produces a violet pigment called violacein that carry anticancer as well anti-parasitic properties. It also produces hydrogen cyanide which can solubilize gold; thus, making it a potential target for metabolic engineering. In silico metabolic modelling utilizing genome-scale models (GSMs) and kinetic models (KMs) offers predictive insights to guide engineering strategies. Although GSMs are more widely implemented, KMs provide a superior, dynamic view of metabolic networks and allow for a more intuitive integration of simulated genetic modifications. Although a GSM of *C. violaceum* has been published, there had been no KM of *C. violaceum* to date. Therefore, this study aims to develop a whole-cell KM of *C. violaceum* ATCC 12472, The resulting KM, cviFA26, consists of 618 distinct enzyme commission (EC) numbers linked to 1736 enzymatic reactions involving 1255 metabolites. This serves as a foundational KM for future development.

Keywords: Whole-Cell Model; Kinetic Model; Opportunistic Pathogen; Violacein; Hydrogen Cyanide; Differential Equations; AdvanceSyn Toolkit

Introduction

Chromobacterium violaceum ATCC 12472 is a Gram-negative beta-proteobacterium [1] found mainly in tropical and subtropical areas [2]. Though rare in reported cases globally, *C. violaceum* ATCC 12472 is known as an opportunistic pathogen that causes severe infection once entering the human body [3], with about 60% mortality rate despite antibiotic treatment [4]. However, it also

produces violacein, a strong antimicrobial agent [5] with potential anticancer properties [6]. Violacein, being violet in colour, is also considered for potential use as a dye [7]. *C. violaceum* ATCC 12472 also produces hydrogen cyanide which can solubilize gold via biocyanidation [8,9]. This makes *C. violaceum* a potential target for metabolic engineering [10,11].

Metabolic engineering relies heavily on modelling frameworks that can simulate pathway behaviour and guide genetic interventions [12,13]. GSMs and KMs form the primary options [14,15]. GSMs are useful for reaction-rate analyses but remain limited to flux predictions. KMs extend beyond this by providing both yield and rate outputs [16], and they support gene knock-ins with greater modelling flexibility [17]. These properties make KMs a strong choice for computational comparison of engineering ideas. Consequently, the community has increasingly emphasised the need to develop and expand kinetic model libraries [18,19].

Although a GSM of *C. violaceum*, iDB858, has been reported [20]; there has been no whole-cell KM of *C. violaceum* to date. Hence, this study aims to construct a KM of *C. violaceum* ATCC 12472 using *ab initio* approach by identifying enzymes from its published genome [21], and identifying the corresponding reaction from KEGG [22]. The result is a whole cell KM of *C. violaceum* ATCC 12472, named as cviFA26 using the nomenclature proposed by Cho and Ling [23], which consists of 1255 metabolites, 618 enzymes with corresponding transcriptions and translations, and 1736 enzymatic reactions; and has been deposited into KinModRe [24].

Materials and Methods

Identification of reactome

The complete genome of *Chromobacterium violaceum* ATCC 12472 (NCBI RefSeq assembly GCF_000007705.1; Chromosome RefSeq NC_005085.1) was used as source to identify enzymatic genes using the process previously described [17,25,26]. Annotated genomes often contain information pertaining to enzymatic genes and tagged with relevant Enzyme Commission (EC) numbers; hence, represents a stable source of information to identify enzymatic reactions. For example, each enzymatic gene was identified as a presence of complete EC number in the GenBank record and mapped into reaction IDs via KEGG Ligand Database for Enzyme Nomenclature [22]. For example, EC 1.1.1.23 (<https://www.genome.jp/entry/1.1.1.23>) catalyses reactions R01158, R01163, and R03012; where the substrates and products of each reaction can be identified.

Model development

The model was developed using the ordinary differential equation (ODE) formats described in Sim., *et al.* [27] for enzymatic reactions. However, for transcription and translation of enzymatic genes to enzymes, we employ the same format used in similar

whole cell KM constructions [17,26,28-32]. Briefly, BioNumbers indicate the presence of about 3000 RNA polymerases in *Escherichia coli* (BioNumbers 106199) [33] but only 25% are actively elongating (BioNumbers 111676) [34] at 22 nucleotides per second (BioNumbers 104109) [35]. Each nucleotide weighs 339.5 Da, giving a synthesis rate near 5600 kDa of RNA per second, equivalent to 9.3e-18 grams per second. With a cell volume of 7e-16 litres [36] and 4225 genes (BioNumbers 105443) [37], this equates to 2.92 micromolar per gene per second. Given a transcript life of 107.56 seconds (BioNumbers 107666) [38] (0.93% decay per second), the transcription ODE becomes $d[\text{mRNA}]/dt = 0.00292 - 0.0093[\text{mRNA}]$. Translation proceeds at about 0.278 peptides per second per mRNA (BioNumbers 106382) [39], and proteins degrade slowly at 2.78e-6 per second (BioNumbers 109924) [40], giving $d[\text{peptide}]/dt = 0.278[\text{mRNA}] - 0.00000278[\text{peptide}]$. *E. coli*'s transcription and translation kinetics were used instead of *C. violaceum*'s as they are generally not available in BioNumbers at this point in time. ODEs representing the full reactome [25,41] were parameterized with typical enzyme constants (kcat of 13.7 per second, Km of 1 mM) [42] following AdvanceSyn Model Specification [43].

Model simulation

The constructed model was tested for simulatability using AdvanceSyn Toolkit [43]. Initial concentrations of all mRNA and enzymes were set to 0 mM. Initial concentrations of all metabolites were set to 1 mM except the following which were set to 1000 mM: (I) C00001 (Water), (II) C00002 (ATP), (III) C00003 (NAD⁺), (IV) C00004 (NADH), (V) C00005 (NADPH), (VI) C00006 (NADP⁺), (VII) C00007 (Oxygen), (VIII) C00011 (Carbon Dioxide), (IX) C00014 (Ammonia), (X) C00025 (L-Glutamate), (XI) C00031 (D-Glucose), (XII) C00032 (Heme), (XIII) C00037 (Glycine), (XIV) C00041 (L-Alanine), (XV) C00042 (Succinate), (XVI) C00047 (L-Lysine), (XVII) C00049 (L-Aspartate), (XVIII) C00064 (L-Glutamine), (XIX) C00065 (L-Serine), (XX) C00073 (L-Methionine), (XXI) C00096 (GDP-mannose), (XXII) C00097 (L-Cysteine), (XXIII) C00099 (Beta-Alanine), (XXIV) C00132 (Methanol), (XXV) C00133 (D-Alanine), (XXVI) C00134 (Putrescine), (XXVII) C00135 (L-Histidine), (XXVIII) C00138 (Reduced ferredoxin), (XXIX) C00139 (Oxidized ferredoxin), and (XXX) C00148 (L-Proline). These 30 metabolites were set to 1000 mM to represent abundance in the media rather than absolute concentrations, which is synonymous to setting generic bounds of -1000 to 1000 in GSMs for media components

or any components representing abundance [44,45]. The model was simulated using the fourth-order Runge-Kutta method [46,47] from time zero to 3600 seconds with timestep of 0.1 second, and the concentrations of metabolites were bounded between 0 millimolar and 1000 millimolar. The simulation results were sampled every 2 seconds.

Results and Discussion

The resulting whole-cell KM of *C. violaceum* ATCC 12472 developed using AdvanceSyn Model Specification [43] as cviFA26; consists of 618 enzymes with 1736 enzymatic reactions involving 1255 metabolites, which is 33% more reactions and 9.7% more metabolites than that of iDB858 with 1303 reactions and 1144 metabolites [20]. In addition, 1236 ODEs acting as placeholders for transcription and translation were added.

Using AdvanceSyn Toolkit [43], we executed cviFA26 and observed stable simulation behaviour (Figure 1), confirming that the model is syntactically correct and functionally coherent as argued in recent model constructions [17,26,28-32]. The production of diphosphoric acid and L-citrulline should be regarded as a modeling artefact, as all reactions share the same median kinetic constants drawn from a broad enzyme survey [42]. This choice was intentional to allow structural validation before refined parameterization. However, a recent sensitivity analysis study [48] varying k_{cat} and K_m values suggests that using median k_{cat} and K_m values given by Bar-Even, *et al.* [42] may also provide substantial predictability.

Nevertheless, there are several limitations with this model. Firstly, it uses *E. coli*'s transcription and translation kinetics instead of *C. violaceum*'s, and all promoters are deemed constitutive with identical strength. Secondly, it is the use of uniform enzyme kinetics for enzymatic reactions. However, these may be resolved with future data from *C. violaceum*. Thus, the main output is a fully operable first-draft whole-cell kinetic model of *C. violaceum* ATCC 12472, which others may adopt or extend; for example, adding organism-specific enzyme kinetics [49-51] as the current ODEs act as placeholders, incorporation of additional regulatory layers or as a system to examine cellular resource allocations [52-55].

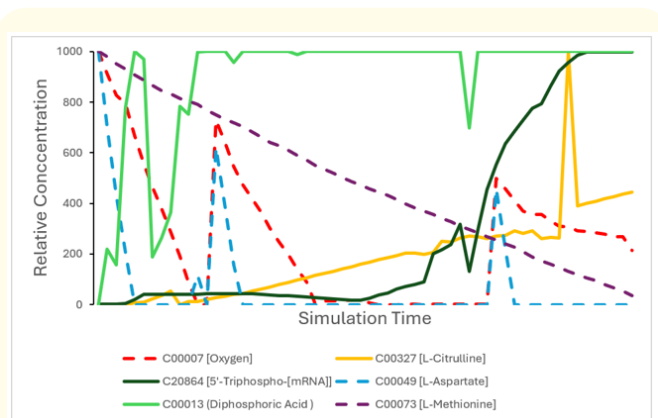


Figure 1: Selection of Simulation Results.

Conclusion

In this study, we present an *ab initio* whole cell kinetic model of *Chromobacterium violaceum* ATCC 12472. The resulting kinetic model, cviFA26; comprising of 1255 metabolites, 618 enzymes with corresponding transcriptions and translations, and 1736 enzymatic reactions.

Supplementary Materials

Reaction descriptions and model can be download from <https://bit.ly/cviFA26>.

Conflict of Interest

The authors declare no conflict of interest.

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